Forecasting Dividend Behavior of Firms Listed on the Colombo Stock Exchange

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Project Final Report

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DECLARATION

I certify that this is my own work, and that this thesis does not incorporate without acknowledgment any material previously submitted for a degree or diploma at any other university or Institute of higher learning, and that it does not includes any material previously released or written by another person, except where acknowledgement is made in the text.

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ABSTRACT

Dividend forecasting plays a crucial role in investment decision-making, particularly in emerging markets such as Sri Lanka, where investors rely heavily on dividend income for portfolio stability and long-term returns; however, predicting dividend behavior in the Colombo Stock Exchange (CSE) is complex due to the combined influence of firm-level financial indicators and macroeconomic factors. Traditional forecasting approaches based on statistical and econometric models are limited in capturing non-linear relationships and often lack practical applicability for investors. This research proposes the development of an AI-driven Dividend Forecasting and Analysis System designed as a decision-support tool to analyze and predict dividend behavior of CSE-listed companies. The system integrates historical dividend data, company financial indicators, and macroeconomic variables to identify patterns and forecast trends such as whether a firm's dividend payout is likely to increase, decrease, or remain stable in the upcoming period. Additionally, the system introduces scenario-based analysis, enabling users to adjust macroeconomic factors such as inflation, interest rates, and GDP growth to evaluate their potential impact on dividend decisions. The proposed solution will be implemented as an interactive web-based dashboard, allowing investors, analysts, and policymakers to explore historical trends, compare companies, and make informed, data-driven investment decisions. By combining machine learning techniques with a user-oriented analytical interface, this research aims to bridge the gap between theoretical financial models and practical decision-making, ultimately enhancing the ability to anticipate dividend behavior within the Sri Lankan capital market.

Keywords: Dividend Forecasting, Machine Learning, Colombo Stock Exchange (CSE), Decision Support System, Financial Analytics, Emerging Markets

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LIST OF ABBREVIATIONS

CSE	Colombo Stock Exchange
ML	Machine Learning
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
XGBoost	Extreme Gradient Boosting
LR	Logistic Regression
GDP	Gross Domestic Product
EPS	Earnings Per Share
ROE	Return on Equity
SDLC	Software Development Life Cycle
WBS	Work Breakdown Structure

INTRODUCTION

General Introduction

Dividends represent a key component of investment returns and serve as an important indicator of a company's financial health and stability. In emerging markets such as Sri Lanka, investors often rely on dividend income as a primary source of return, especially in environments characterized by market volatility and economic uncertainty. Within the Colombo Stock Exchange (CSE), dividend-paying companies are generally perceived as more reliable and attractive, as consistent dividend payments signal profitability, effective financial management, and long-term sustainability.

However, understanding and predicting dividend behavior is a complex task. Dividend decisions are influenced by a combination of internal financial factors such as earnings, liquidity, leverage, and growth opportunities, as well as external macroeconomic conditions including inflation, interest rates, and overall economic performance. These interconnected factors make dividend behavior dynamic and difficult to analyze using simple or traditional approaches. As a result, investors often face challenges in anticipating whether a company's dividend payout will increase, decrease, or remain stable in future periods.

In practice, most investors in the Sri Lankan market depend on manual analysis of financial statements, historical trends, and external reports to make dividend-related decisions. This process is time-consuming, requires significant financial knowledge, and may still lead to inaccurate conclusions due to the complexity of the underlying relationships. Furthermore, existing research in the Sri Lankan context has primarily focused on identifying factors that influence dividend policies rather than developing practical tools that support real-world decision-making.

With advancements in data analytics and machine learning, there is an opportunity to move beyond traditional analysis toward predictive and data-driven solutions. Machine learning techniques are capable of capturing complex, non-linear relationships between financial and macroeconomic variables, making them suitable for analyzing dividend behavior in dynamic market conditions. Leveraging these capabilities, this research proposes the development of an AI-driven decision-support system that assists investors in understanding and forecasting dividend behavior of CSE-listed companies.

The proposed system integrates historical dividend data, company financial indicators, and macroeconomic variables to provide insights into future dividend trends. Through an interactive web-based dashboard, users will be able to explore patterns, analyze influencing factors, and evaluate how changes in economic conditions may impact dividend decisions. By focusing on practical usability

and real-world application, this research aims to enhance investor decision-making and contribute to a more data-driven investment environment within the Sri Lankan capital market.

Background literature

Dividend policy has long been a central topic in corporate finance, as it reflects how companies distribute earnings to shareholders while balancing reinvestment needs. Several classical theories provide a foundation for understanding dividend behavior. The Dividend Irrelevance Theory suggests that dividend decisions do not affect firm value under perfect market conditions, while the Signaling Theory proposes that dividends convey information about a firm's future prospects. Agency Theory highlights dividends as a mechanism to reduce conflicts between managers and shareholders, and the Bird-in-the-Hand Theory argues that investors prefer certain dividend income over uncertain capital gains.

Although these theories offer valuable insights, their assumptions often do not hold in real-world contexts, particularly in emerging markets such as Sri Lanka. In such environments, dividend decisions are influenced not only by firm-level financial performance but also by broader economic conditions, regulatory frameworks, and market uncertainties. Factors such as profitability, liquidity, leverage, and firm size, along with macroeconomic variables like inflation, interest rates, and economic growth, collectively shape dividend behavior. As a result, dividend policy in the Colombo Stock Exchange (CSE) reflects a complex interaction of internal and external influences rather than a single theoretical perspective.

Despite the importance of dividends in investment decision-making, existing approaches in Sri Lanka largely focus on descriptive and explanatory analysis. These approaches identify relationships between variables but provide limited support for anticipating future dividend behavior. This creates a need for more advanced and practical methods that can assist investors in understanding how dividend patterns may evolve over time.

A considerable body of research has examined dividend policy and its determinants both in Sri Lanka and globally. In the Sri Lankan context, empirical studies have identified key factors influencing dividend decisions, including profitability, earnings stability, liquidity, and ownership structure. These studies primarily rely on econometric models such as regression analysis to explain how these variables affect dividend payouts. While such findings contribute to theoretical understanding, they

remain limited in their ability to provide forward-looking insights or support real-time investment decisions.

In contrast, recent global research has explored the application of machine learning techniques in financial forecasting, including dividend behavior analysis. Machine learning models such as Random Forest, gradient boosting algorithms, and neural networks have demonstrated improved performance over traditional statistical methods due to their ability to capture complex and non-linear relationships among variables. Studies conducted in various emerging markets have shown that these models can effectively classify dividend-related outcomes and identify patterns that are not easily detected through conventional approaches.

However, despite these advancements, several limitations remain. Many machine learning-based studies focus primarily on classification tasks, such as predicting whether a company will pay dividends, without extending their findings into practical tools for investors. Furthermore, most existing research does not adequately incorporate macroeconomic scenario analysis, which is particularly important in volatile economic environments like Sri Lanka. The absence of user-oriented systems means that these models are often confined to academic settings and are not accessible for real-world decision-making.

Additionally, there is a noticeable lack of research specifically tailored to the Colombo Stock Exchange that integrates financial indicators, macroeconomic variables, and predictive analytics into a unified framework. Existing solutions do not provide interactive capabilities that allow users to explore how changes in economic conditions may influence dividend behavior.

Therefore, while prior studies offer valuable theoretical and methodological contributions, there remains a clear gap in translating these insights into practical, decision-support systems. This research aims to address this gap by developing an AI-driven platform that not only analyzes dividend behavior but also enables users to make informed predictions about future dividend trends in a dynamic and interactive manner.

Research Gap

Despite the extensive body of research on dividend policy, several important limitations can be identified, particularly within the context of the Colombo Stock Exchange (CSE). Existing studies in Sri Lanka primarily focus on identifying the determinants of dividend decisions using traditional econometric approaches. While these studies provide valuable insights into the factors influencing dividend behavior, they are largely descriptive in nature and do not offer predictive capabilities that can support forward-looking investment decisions.

Furthermore, although global research has increasingly adopted machine learning techniques for financial forecasting, these approaches are often limited to experimental or academic settings. Many of these models focus on isolated prediction tasks without being integrated into practical, user-oriented systems. As a result, there is a lack of accessible tools that translate analytical findings into actionable insights for investors.

Another significant limitation is the insufficient incorporation of macroeconomic variables in predictive analysis. Dividend behavior in emerging markets is highly sensitive to economic conditions such as inflation, interest rates, and economic growth; however, many existing models do not provide mechanisms to evaluate how changes in these factors may influence future dividend trends.

Additionally, there is no comprehensive system specifically designed for the Sri Lankan market that combines historical dividend data, firm-level financial indicators, and macroeconomic variables into a unified analytical framework. The absence of interactive platforms further limits the ability of investors to explore patterns, compare companies, and assess potential future outcomes in a structured manner.

Therefore, a clear research gap exists in the development of a practical, data-driven decision-support system that enables users to analyze and anticipate dividend behavior within the Colombo Stock Exchange.

RESEARCH PROBLEM

Dividend behavior plays a critical role in investment decision-making, yet accurately anticipating changes in dividend patterns remains a significant challenge in the Colombo Stock Exchange (CSE). Investors often depend on manual analysis of financial statements, historical trends, and external reports to assess whether a company's dividend payout is likely to increase, decrease, or remain stable. This process is not only time-consuming but also requires a high level of financial expertise, and it may still lead to inaccurate conclusions due to the complexity and dynamic nature of influencing factors.

Existing approaches in Sri Lanka are predominantly based on traditional statistical methods, which are limited in their ability to capture non-linear relationships between financial and macroeconomic variables. As a result, these methods are not well-suited for predicting dividend behavior in a rapidly changing economic environment. Moreover, while machine learning techniques have shown potential in improving prediction accuracy, their application in the Sri Lankan context remains limited, and they are rarely implemented as practical tools for end users.

Another key issue is the lack of integration between financial data and macroeconomic conditions when analyzing dividend trends. Investors currently do not have access to systems that allow them to evaluate how changes in economic factors, such as inflation or interest rates, may impact future dividend decisions. This lack of comprehensive and interactive analysis further complicates the decision-making process.

Consequently, there is a need for a system that can bridge the gap between complex financial data and practical investment decision-making. Specifically, a solution is required to support investors in understanding and predicting dividend behavior in a structured, data-driven manner. This research therefore addresses the problem of how to develop an effective decision-support system that leverages machine learning techniques to analyze historical patterns, incorporate macroeconomic factors, and provide meaningful insights into the future dividend behavior of CSE-listed companies.